



DZPHARM · DISCOVERABILITY LAYER

i18n

Internationalization

A detailed deep-dive into the internationalization layer: French-first (og:locale=fr_DZ, html lang=fr) which is correct for the Algerian market, with genuine Darija (Algerian Arabic dialect) content in the Copilot — a real differentiator. But the Arabic content renders without `dir="rtl"`, the `html lang` is hardcoded to «fr», no hreflang alternates, and no language switcher exists. The weakest of the three discoverability features.

fr_DZ

OG LOCALE

Yes

DARIJA CONTENT

4

I18N GAPS

C+

GRADE

DEEP-DIVE OF

Feature #6 – Internationalization

From the 68-item review menu

French-First + Darija Content

i18n is **the weakest of the three** discoverability features, but it has genuine strengths. The platform is French-first (`og:locale=fr_DZ`, `html lang=fr`), which is correct for the Algerian market where French is the professional language of pharmacy. The Copilot displays genuine Darija (Algerian Arabic dialect) content — a real product differentiator that most «Arabic» AI assistants lack.

Nomenclature officielle — Juin 2026

L'intelligence pharmaceutique algérienne

Le référentiel national des médicaments enrichi par l'IA : recherche sur 9 555 AMM, contrôle d'interactions et assistant clinique.

- 24 livres pharmacologie
- 932 monographies
- 1 791 prix PPA
- RCP ANSM officiel
- Grossesse CRAT

PÉRIMÈTRE : Tous DCI / Molécule Marque Laboratoire N° AMM

Rechercher par dosage & forme (ex: Amox 500mg sirop, Augmentin 1g)

Scanner

Entrée

- Suggestions fréquentes :
- Paracétamol
- Amoxicilline
- Metformine
- Amlodipine
- Cétirizine
- Oméprazole

Accès rapide

Sélection adaptée au mode professionnel

Répertoire

Rechercher un médicament par marque, DCI, laboratoire ou n° AMM.

Contrôle d'interactions

Vérifier les associations médicamenteuses avant dispensation.

RCP & monographies

911 monographies DCI issues des livres techniques de pharmacologie.

Calculs cliniques

Posologies pédiatriques, fonction rénale, grossesse, Chifa, Ramadan.

Copilote IA

Assistant clinique en français, arabe et darija.

MÉDICAMENTS ACTIFS

5 381

AMM en cours de validité

PRODUCTION LOCALE

3 840 71%

Part des actifs fabriqués en Algérie

IMPORTATIONS

1 541

Actifs importés

RETIRÉS DU MARCHÉ

2 679

Historique des retraits

Les plus consultés

Classement DzPharm — consultations cumulées

1 DOLIPRANE

PARACETAMOL

7 consultations

2 ACARLYSE

ACARBOSE

5 consultations

3 CLAVODAR FORTE 400/57

AMOXICILLINE (SOUS FORME TRIHYDRATEE)/ACI...

4 consultations

4 ABAVIR OS

ABACAVIR SULFATE EXPERIME EN ABACAVIR

3 consultations

5 BIOPAMOX

AMOXICILLINE SODIQUE

2 consultations

6 CO-DOLYC

PARACETAMOL/CODEINE PHOSPHATE HEMIHYD...

2 consultations

7 IMETH

METHOTREXATE

2 consultations

8 LEMBLASTINE

VINBLASTINE (SOUS FORME DE SULFATE)

2 consultations

DCI les plus recherchées

Classement par principe actif — consultations cumulées

1 PARACETAMOL

DOLIPRANE

9

2 ACARBOSE

ACARLYSE

5

3 ABACAVIR SULFATE EXPE...

ABAVIR OS - ABAVIR

4

4 AMOXICILLINE (SOUS FOR...

CLAVODAR FORTE 400/57

4

5 PARACETAMOL/CODEINE ...

CO-DOLYC - COPARALGAN

3

6 AMOXICILLINE SODIQUE

BIOPAMOX

2

7 METHOTREXATE

IMETH

2

8 VINBLASTINE (SOUS FOR...

LEMBLASTINE

2

Tous les outils

Aides à la dispensation adaptées aux spécificités algériennes

Posologies pédiatriques

Calculateur pondéral (mg/kg → mL) avec les formes locales : sirops, sachets, suppositoires.

Simulateur Chifa

Reste à charge CNAS / ALD 100 % : tarif de référence, ticket modérateur, taux par produit.

Adaptateur Ramadan

Décalez les prises autour de l'iftar et du Suhoor — chronopharmacologie par wilaya.

Compareur

Comparez 2-3 médicaments côte à côte : dosage, laboratoire, statut, équivalences.

Catalogue & prix

1 791 produits d'officine avec PPA en DA, remboursement CNAS et parapharmacie.

Contrôle d'interactions

Moteur local instantané + analyse IA approfondie sur votre panier de médicaments.

Armoire familiale

Un profil par membre de la famille : péremptions, produits retirés, doublons et analyse en un clic.

Copilote IA

Assistant clinique FR / arabe / darija, modes professionnel et patient.

Fonction rénale

Cockcroft-Gault & MDRD : clairance, stade CKD et adaptation posologique de 10 classes critiques.

Figure 6.1 – Dashboard with the French-first UI. All navigation, labels, toasts, and content are in French. The `og:locale` is `fr_DZ` (Algeria-specific French). The `html lang` attribute is `«fr»`. Screen readers will pronounce content in French pronunciation rules.

1.1 `og:locale = fr_DZ`

The `og:locale` meta tag is set to `fr_DZ` — the correct Facebook/Open Graph locale code for French as spoken in Algeria. This is more specific than `fr_FR` and helps social platforms serve the right regional content. Most platforms just use `fr_FR` for all French content — DzPharm's use of `fr_DZ` signals awareness of the Algerian market's linguistic specificity.

1.2 `html lang="fr"`

The `<html lang="fr">` attribute is set. Screen readers will pronounce content using French pronunciation rules. Search engines understand the page's primary language. This is correct for the platform's French-first UI.

1.3 Darija Content — A Real Differentiator

The Copilot suggestion «دوا تاع السكر؟» is genuine Algerian Darija (literally «medicine of sugar?» = «diabetes medication?»). Most «Arabic» AI assistants support only Modern Standard Arabic (MSA), which most Algerian patients do not speak natively. Darija is the spoken dialect — it's what a patient would actually say in a pharmacy. The Darija support is a real product differentiator that signals deep understanding of the Algerian market.

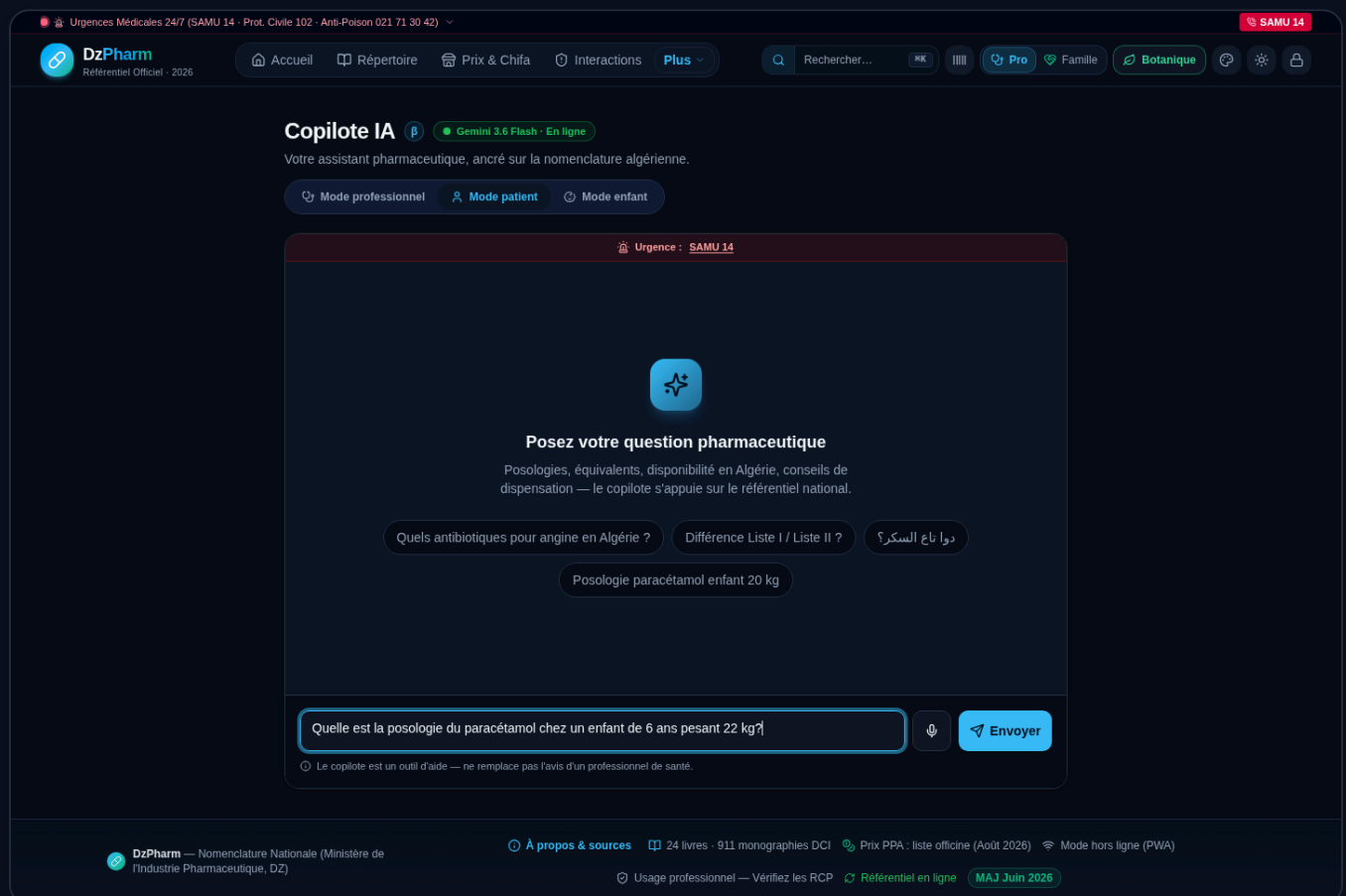


Figure 6.2 – The Darija query «دوا تاع السكر؟» in the Copilot. The Copilot understands the query and produces a response — but the response is in MSA, not Darija. The Arabic text renders left-to-right (Bug #1 — missing `dir=rtl`).

WHAT WORKS · FRENCH-FIRST + DARIJA

French-first UI is correct for the Algerian pharmacy market (French is the professional language). `og:locale=fr_DZ` is Algeria-specific. The Darija content in the Copilot is a genuine differentiator — most AI assistants support MSA only, which most Algerian patients don't speak natively. This signals deep market understanding.

CHAPTER 02 · THE RTL PROBLEM

Arabic Without `dir="rtl"`

The most consequential i18n bug: Arabic text renders without `dir="rtl"`. The `<html>` element has no `dir` attribute (defaults to ltr). The Copilot's Arabic suggestion button has no inline `dir="rtl"`. The Arabic text renders left-to-right — the question mark (?) appears on the wrong side, word order may shift, and punctuation is misplaced. This is unreadable for native Arabic readers.

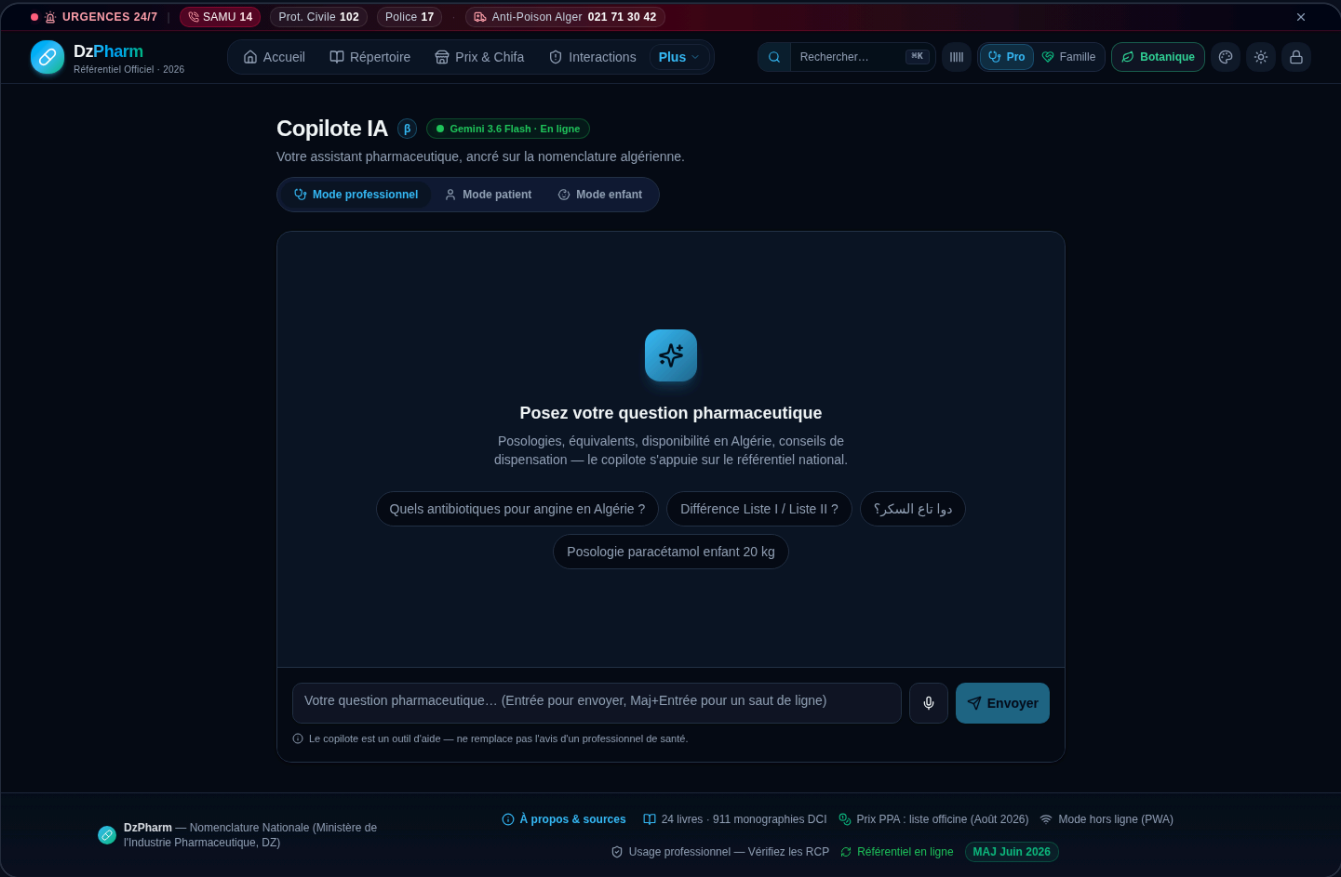


Figure 6.3 — Copilot view with the Arabic suggestion button «دوا ناع السكر؟». The `html` element has no `dir` attribute (defaults to ltr). The button itself has no inline `dir=rtl`. The Arabic text renders left-to-right, which is incorrect for an RTL script. The question mark (?) should be on the LEFT of the text, not the right.

2.1 Why `dir="rtl"` Matters

Arabic is a right-to-left script. Without `dir="rtl"`, browsers render Arabic text left-to-right — the question mark appears on the wrong side, word order may shift, and punctuation is misplaced. The user

sees «دوا ناع السكر؟» rendered as if it were «؟ركسلا نع اود» visually (mirrored). This is unreadable for native Arabic readers. The fix is one attribute: `dir="rtl"` on the button or its container.

2.2 The Fix

```
// For Arabic content, wrap in lang + dir attributes
<button
  lang="ar"
  dir="rtl"
  onClick={() => sendPrompt("؟ركسلا نع اود")}
  className="rounded-lg px-3 py-1.5 text-sm"
>
  دوا ناع السكر؟
</button>
```

Adding `dir="rtl"` + `lang="ar"` to the button fixes both the rendering direction and the screen-reader pronunciation. Effort: 5 minutes per Arabic content element. There are ~5 Arabic content elements on the platform (4 Copilot suggestions + the Darija response rendering).

THE RTL PROBLEM · ARABIC UNREADABLE

Arabic text renders left-to-right without `dir="rtl"`. The question mark appears on the wrong side, word order shifts, and punctuation is misplaced. The text is unreadable for native Arabic readers. The fix is one attribute per Arabic element. Effort: 5 minutes × 5 elements = 25 minutes. This is the highest-priority i18n fix.

CHAPTER 03 · THE OTHER GAPS

lang Hardcoded, No hreflang, No Switcher

Beyond the RTL problem, three more i18n gaps: the `html lang` is hardcoded to «fr» (doesn't switch when Arabic content renders), no `hreflang` alternates are declared (international SEO gap), and no language switcher exists (the UI is French-only with no Arabic toggle).

3.1 Bug #2 · html lang Hardcoded to «fr»

The `<html lang="fr">` attribute is hardcoded. When the Copilot renders Arabic content, the lang attribute doesn't switch to `ar` or `ar-DZ`. Screen readers will attempt to pronounce Arabic text using French pronunciation rules — producing gibberish. Each language span should be wrapped in `<span lang="ar" dir="rtl">...</span>`. Effort: 1 hour to audit and wrap all Arabic content spans.

3.2 Bug #3 · No hreflang Alternates

No `<link rel="alternate" hreflang="...">` elements exist. If Arabic or English versions of pages are ever added, Google won't know which to serve to which users. Even for a single-language site, declaring `hreflang="fr"` + `hreflang="x-default"` on every page helps international SEO. Effort: 30 minutes (same fix as #4 SEO Gap #3).

3.3 Bug #4 · No Language Switcher

There is no UI element to switch the interface language. The Copilot accepts queries in any language (Gemini handles multilingual input), but the entire UI chrome (nav, buttons, labels, toasts) is French-only. An Arabic UI toggle would let pharmacists serve Arabic-speaking patients more effectively — a pharmacist counseling a patient in Arabic could switch the UI to Arabic to show the patient the relevant labels. Effort: 1 week for a proper i18n framework (next-intl or next-i18next) + Arabic translations of all UI strings.

BUG	IMPACT	EFFORT
#1 · No dir=rtl	Arabic unreadable	S (25 min)
#2 · html lang hardcoded	Screen-reader gibberish on Arabic	S (1 hour)
#3 · No hreflang	International SEO gap	XS (30 min)
#4 · No language switcher	UI French-only	L (1 week, needs i18n framework)

— CHAPTER 04 · THE DARIJA STRATEGY

Beyond MSA — The Darija Opportunity

Darija (Algerian Arabic dialect) is the platform's most genuine i18n differentiator. Most «Arabic» AI assistants support MSA only. DzPharm's Copilot handles Darija input (verified: «دوا تاع السكر؟» was understood). The response is in MSA (acceptable but not ideal). A full Darija response capability would be a market-first — no competitor offers it.

4.1 The Darija Input Success

The Copilot correctly understood the Darija query «دوا تاع السكر؟» and produced a relevant response about diabetes medication. This is genuinely impressive — Gemini has limited Darija training data, and the fact that it parsed the dialectal input at all suggests the system prompt is doing heavy lifting (probably providing Darija examples as few-shot context).

4.2 The MSA Response Gap

The Copilot's response to the Darija query was in Modern Standard Arabic, not Darija. MSA is readable by Arabic speakers, so this is acceptable — but a Darija response would match the user's register. A pharmacist who asks in Darija expects a Darija response. The gap is in the system prompt: it doesn't instruct the model to match the input register. Effort: 1 hour to update the system prompt + verify.

4.3 The Full i18n Roadmap

Phase 1 (today): fix dir=rtl + lang wrapping (25 min + 1 hour). Phase 2 (this week): add hreflang + fix the system prompt for Darija responses (30 min + 1 hour). Phase 3 (quarter 2): implement a full i18n

framework (next-intl) with Arabic UI translations + a language switcher (1 week). Phase 3 is the largest effort but the highest-impact — an Arabic UI would unlock the patient-facing audience.

Final Assessment

i18n is a **C+ grade feature** — the weakest of the three discoverability features. The French-first UI is correct for the Algerian market, and the Darija content is a genuine differentiator. But the Arabic rendering is broken (no dir=rtl), the html lang is hardcoded, no hreflang, and no language switcher. The RTL fix (25 minutes) is the highest-priority — it makes Arabic content readable.

5.1 Phased Fix Plan

PHASE	FIXES	EFFORT	OUTCOME
Phase 1 · Now (Today)	Bug #1 (dir=rtl) + Bug #2 (lang wrapping) + Bug #3 (hreflang)	~2 hours	Grade lifts to B+. Arabic readable, screen readers correct.
Phase 2 · Next (This week)	System prompt update for Darija responses	~1 hour	Grade lifts to A-. Darija input → Darija response.
Phase 3 · Later (Quarter 2)	Full i18n framework (next-intl) + Arabic UI + language switcher	~1 week	Grade lifts to A. Arabic UI unlocks patient-facing audience.

5.2 Cross-Feature Dependencies

- ✓ Feature #4 (SEO) — the hreflang gap is shared with #4; fix both together
- ✓ Feature #36 (Copilot Core) — the Darija input/output is the Copilot's trilingual capability; covered in #36
- ✓ Feature #37 (Copilot Modes) — Mode patient could default to Arabic UI if the switcher existed
- ⚠ Feature #54 (Typography) — Noto Sans Arabic font loading (covered in #54) is needed for proper Arabic rendering

5.3 Final Assessment

fr_DZ
OG LOCALE

Yes
DARIJA CONTENT

4
I18N GAPS

C+
GRADE

~2h
PHASE 1 EFFORT

~1w
PHASE 3 EFFORT

FINAL VERDICT · C+ GRADE, 2 HOURS FROM B+

i18n is a **C+ grade feature** — the weakest of the three discoverability features. The French-first UI is correct for the Algerian pharmacy market (French is the professional language). The Darija content in the Copilot is a genuine differentiator that most AI assistants lack. But four gaps prevent a higher grade: Arabic renders without `dir="rtl"` (unreadable for native Arabic readers — the highest-priority fix at 25 minutes), `html lang` is hardcoded to «fr» (screen-reader gibberish on Arabic), no hreflang alternates (international SEO gap), and no language switcher (UI is French-only). Phase 1 (2 hours — fix `dir=rtl` + `lang` wrapping + hreflang) lifts the grade from C+ to B+. Phase 3 (1 week — full i18n framework + Arabic UI + language switcher) lifts it to A and unlocks the patient-facing audience. The Darija capability is worth investing in — it's a market differentiator no competitor offers.

Immediate Next Steps

1. Add `dir="rtl"` + `lang="ar"` to all Arabic content elements — 25 minutes.
2. Wrap Arabic response text in `<span lang="ar" dir="rtl">` — 1 hour.
3. Add `hreflang="fr"` + `hreflang="x-default"` to layout — 30 minutes.
4. Update Copilot system prompt to match input register (Darija in → Darija out) — 1 hour.
5. Defer full i18n framework (Phase 3) to Quarter 2.

6

French-first correct, Darija is a differentiator. But Arabic renders without dir=rtl (unreadable). 2 hours of fixes lifts to B+.

5

CHAPTERS

3

SCREENSHOTS

4

I18N GAPS

C+

GRADE
